

Channel Correlation: the Connected Correlation is $\ln \gcd$

The first relational quantity, and it lands on the resistance isomorphism’s lattice core

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Framing. The catalog mapped single operations, each an $\ln(\mathbf{fan-in})$ — a single-integer resistance. The first *relational* question — how much do two modulo channels share — produces the first two-variable quantity. Erasure-only discipline, claims graded Fact / Reading / Lead, verified by `channel_correlation.py`.

1. The result

Run $\mathbf{a} \bmod m$ and $\mathbf{a} \bmod m$ on the same $\mathbf{a}$, uniform over one full joint period $N = \text{lcm}(m, m)$. Then

$$I(\mathbf{a} \bmod m_1; \mathbf{a} \bmod m_2) = \ln \gcd(m_1, m_2),$$

confirmed numerically across all pairs 2..8 to machine precision ($\max |I - \ln \gcd| = 2 \times 10^{-1}$). The Chinese Remainder structure makes it exact: both channels determine $\mathbf{a} \bmod \gcd$ and nothing more, and $I = \ln m + \ln m - \ln \text{lcm} = \ln \gcd$. (*Fact.*)

2. Coprimality is zero correlation

The $\gcd$ matrix is the coprimality lattice; the correlation is $\ln$ of each cell. Coprime pairs carry $I = 0$. **Two number-channels share information iff they share a prime**, and exactly $\ln(\text{shared part})$ of it. The multiplicative core has become a thermodynamic signal. (*Fact.*)

3. It is the $\Omega_{\gcd}$ term of the resistance isomorphism

$\ln \gcd$ is still a resistance value — the resistance of the integer $\gcd$ — so this is not an escape from $\ln(\text{integer})$. It is sharper than that: it is $\Omega_{\gcd}$, the subtracted correlation term in Paper 1’s union law

$$\Omega_{\text{lcm}} = \Omega_a + \Omega_b - \Omega_{\gcd}.$$

The mutual information of two modulo channels landed exactly on that term. So the relational question reached the **lattice core** of the resistance isomorphism (the $\gcd/\text{lcm}$ structure), not a single-integer rung. The project’s *Innovation = Connected Correlation* finding now has its arithmetic face: the connected correlation between two number-channels is $\ln \gcd$. (*Identification with $\Omega_{\gcd}$: Fact. “Relational core, not single-integer ladder”: Reading.*)

4. The honest boundary, and the direction it opens

Coprimality *emerges* as zero correlation but is not *derived*: m , m are still inputs, so this is correlation-as-divisibility-test, not primes-from-nothing — the same standing problem the modulo gateway named. What is new is that the test is now *relational*: it reads the shared prime structure of a pair, which single-integer resistance cannot express, and the whole coprimality lattice ($6/\pi^2$ density of coprime pairs, Paper 1 Appendix A) is reproduced as a correlation pattern.

Next-direction flag (deep, speculative, later). The resistance isomorphism's zeta reading (Paper 8) is already a superposition/interference story: each integer is a phasor of amplitude $n^{-1/2}$ and phase $-t \ln n$, the nontrivial zeros are total destructive interference of the whole phasor bouquet on the critical line $\sigma = 1/2$, and the amplitude-decay and phase rates are locked by $-i$ (Cauchy–Riemann). That $-i$ is the same i we *generated* discretely from succession and the probe ($[S,D] = -i_y$ at the genesis bit). The avenue: whether the continuous critical-line cancellation is the shadow of a discrete proto-phase structure, with the generated i as the bridge. Held as a direction only — the corpus is explicit that the zeta work gives no leverage on the location of the zeros, and this claims none. (*Lead.*)

Originating direction (the gcd as native to information systems and central to the resistance isomorphism; the relational channel-correlation test; the proto-phase / critical-line connection as a later thread) is Alexander Pisani's; the test and this compilation were developed jointly with Claude (Anthropic). June 2026.